

The Effect of Expected Income on Individual Migration Decisions

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Roadmap

- ① Motivation & Stylized Facts
- ② Model Intuition
- ③ Estimation Idea
- ④ Results: What Drives Migration?
- ⑤ Interpretation & Policy Angles
- ⑥ Q&A

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- 6 Conclusion

Question & Motivation

- **Question:** Do expected income differences drive interstate migration?
- Migration is common for the young; repeat and return moves are pervasive.
- Policy relevance: local shocks, subsidies, welfare differentials.

Stylized Facts (NLSY79)

TABLE I
INTERSTATE MIGRATION, NLSY 1979–1994^a

	Less Than High School	High School	Some College	College	Total
Number of people	322	919	758	685	2,684
Movers (age 18–27)	80	223	224	341	868
Movers (%)	24.8%	24.3%	29.6%	49.8%	32.3%
Moves per mover	2.10	1.95	1.90	2.02	1.98
Repeat moves (% of all moves)	52.4%	48.7%	47.4%	50.5%	49.5%
<i>Return migration</i> (% of all moves)					
Home	32.7%	33.1%	29.1%	23.2%	28.1%
Not home	15.5%	7.1%	6.8%	8.6%	8.4%
Movers who return home	61.3%	56.5%	51.3%	42.8%	50.2%

^aThe sample includes respondents from the cross-section sample of the NLSY79 who were continuously interviewed from ages 18 to 28 and who never served in the military. The home location is the state of residence at age 14.

- Repeat moves are about half of all moves; many are returns to origin.
- Movers are younger; return migration is substantial.

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Model in One Slide

- **Setup:** Locations differ in mean wages and match quality; moving is costly.
- **Information:** Current location wage known; other locations' matches known only by visiting (experience good).
- **Decision:** Maximize expected lifetime income net of moving costs.

- Return migration and onward moves require dynamics.
- Many alternatives matter: returning is a choice among many states.

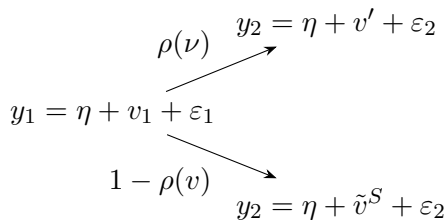
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Computational Challenge → Limited-History Approximation

- Full state space explodes with locations and match histories.
- **Idea:** Keep only the most recent M locations for wages/matches.
- Preserves behavior seen in the data while making DP feasible.



- with probability $\rho(v)$, move and get observed wage $y_2 = \eta + v' + \varepsilon_2$
- with probability $1 - \rho(v)$, stay and get $y_2 = \eta + \tilde{v}^S + \varepsilon_2$ in which $\tilde{v}^S$ is the censored random wage derived by discarding the realized wages of movers

- Mean wage by location; permanent match component; person FE; transitory shocks.
- Migration choice depends on *differences* in expected income and moving cost, not levels.

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- **Expected income matters:** moves respond to geographic mean wages and bad current matches.
- **Costs matter:** distance penalty; strong home premium; cheaper to return.
- **Population size** of destination reduces cost (network/proximity channels).

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- Moving cost equivalent $\sim$ \$300k (PV terms, order of magnitude).
- Good match + high-mean state $\Rightarrow$ $\sim$ \$17k/year gain.
- Migration subsidy thought experiment: interstate rate $\sim$ 2.9% $\rightarrow$ $\sim$ 4.9%.

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- Model reproduces repeat/return migration shares qualitatively.
- Negative association: higher realized income this period $\Rightarrow$ lower out-migration probability.

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What We Learn

- Migration resembles optimal search over locations.
- Both mean differences and idiosyncratic match quality drive moves.
- Structural approach enables policy counterfactuals.

Policy Angles & Extensions

- Local labor demand shocks & wage equalization dynamics.
- Subsidies or relocation assistance: who moves more?
- Household joint decisions; welfare benefit differentials; retirement migration.

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Takeaways

- Expected income plays a sizable role in interstate migration.
- High moving costs + payoff shocks reconcile low average move rates with meaningful gains.
- Model provides a tractable template for counterfactual policy analysis.

Thanks!

Questions? tate.mason@uga.edu

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7 Backup

- NLSY79 cross-section; white non-Hispanic HS grads; ages 20+; not in military.
- Attrition and consistency checks; AFQT use; state means from Census PUMS (1990).

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Likelihood & Identification Sketch (*Backup*)

- Joint likelihood of income & choices; mixture over unobserved components.
- Nonparametric intuition: selection on bad current draws $\uparrow$ move probability.

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Robustness (*Backup*)

- Vary β , horizon T , include preference match, extra amenities.
- Fit diagnostics; simulated vs. observed migration rates.